

TECHNOLOGICAL UNIVERSITY DUBLIN

CITY CAMPUS - GRANGEGORMAN

TU082 BSc. (Honours) Degree in Information Systems /
Information Technology (Part-Time)

Year 3

SEMESTER 1 EXAMINATIONS 2025/26

CMPU3001, Advanced Computer Networking

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Exam Duration: 2 hours.

Attempt Question 1 and three of the four remaining questions. All questions carry equal marks.

1. Figure 1 (A snippet of *daytime* server code) reveals some code for a *daytime* server with error checking code removed for brevity. Refer to this figure and to the following list of terms:

servAddr, inet_ntop, > 0 (the number zero), 2, 3, sockaddr, clntAddr, sin_port, sock, htons, inet_pton, == 0 (the number zero), AF_INET, atoi, servPort, clntSock, sockaddr_in

```
19 char sendbuffer[BUFSIZE];
20
21 in_port_t servPort = XXXXX(argv[1]);
22
23 if (argc != XXXXX)
24
25 int servSock;
26
27 if ((servSock = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP)) < 0)
28
29 if (bind(servSock, (struct XXXXX*) &servAddr, sizeof(servAddr)) < 0)
30
31 if (connect(sock, (struct sockaddr *) &servAddr, sizeof(servAddr)) < 0)
32
33 struct XXXXX servAddr;
34 servAddr.sin_family = XXXXX;
35 servAddr.sin_addr.s_addr = htonl(INADDR_ANY);
36 servAddr.sin_port = htons(XXXXX);
37 memset(&servAddr, 0, sizeof(servAddr));
38
```

Figure 1: A snippet of *daytime* server code.

- (a) There are some errors introduced into this code relating to inappropriate and misplaced code. Using the line numbers provided, identify lines of code that should be removed and, the correct location for any misplaced code. In your answer use terms similar to: “line X should be placed after line Y”, “lines 1&2 should be switched with line 5”, “line Z should be inserted between lines 7&8” etc. Give reasons for your selections. (8 marks)
- (b) Using the line numbers provided, identify which XXXXX value can be replaced with which term from the above list. (12 marks)
- (c) Refer to line 19, explain why a receive buffer is not declared here. (5 marks)

2. In relation to *byte order*:

- (a) Consider a two-byte value to be stored on a computer system. Explain what is meant by **host byte order** identifying the two conventions in use. (8 marks)
- (b) Explain why byte order is particularly significant in relation to segment exchange between TCP entities at the Transport layer. In your answer, focus on addressing data and identify what byte order, if any, TCP uses. (7 marks)
- (c) Explain the purpose of each of the following byte order functions, identifying which direction of transfer each is associated with: Application-to-TCP or, TCP-to-Application:

htons(), *ntohs()*, *inet_pton()* and, *inet_ntop()*

(10 marks)

3. In relation to the concept of a socket, answer the following questions:

- (a) In simplistic terms, explain the purpose of a socket that is not yet associated with a connection. In your answer, identify the relevant network protocol layers that the socket relates to. (4 marks)
- (b) Identify and explain the purpose of all primitives involved in the creation and preparation of a socket (within each of the client and server applications) in readiness for establishing a connection. (4 marks)
- (c) Identify how socket are referenced (identifiers) within the relevant layers of the protocol stack. (6 marks)
- (d) Separately identify the additional socket primitives used within each of the client and server applications to connect sockets. In your answer, explain any differences in the identifiers in (c) above to those used for the connection/connected sockets within the relevant layers of the protocol stack. (11 marks)

4. The following code snippet relates to the *recv()* primitive call in the Berkeley Socket API:

```
while ((numBytes = recv(clntSock, recvbuffer, BUFSIZE - 1, 0)) > 0) {  
    Do something;  
}  
  
if (numBytes < 0)  
    DieWithSystemMessage("recv() failed");
```

- (a) Explain the purpose of this primitive call identifying the protocol layers impacted by the primitive's actions. (6 marks)
- (b) Explain why the primitive call is included in a while loop. In your answer, identifying any factors that might impact on the flow of data across the connection. (6 marks)
- (c) Identify the specific return value that will ensure a graceful exit from the *recv-while* loop and explain what this return value means in relation to the connection. (4 marks)
- (d) Explain why this version of the *recv-while* loop exits gracefully within a **daytime** client application but not within an **echo** server application. In your answer, explain how to overcome this problem within the echo server application. (9 marks)

5. Refer to Figure 2 (*Screenshot from Wireshark*). This screenshot shows a sequence of segment exchanges between hosts hosting a *http* client and *http* server application. Answer the following questions in relation to TCP Data Exchange and the TCP Window mechanism.

1. 56560 → 80 [SYN] Seq=0 Win=65535 Len=0
2. 80 → 56560 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
3. 56560 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
4. 56560 → 80 GET /index.htm HTTP/1.1
5. 80 → 56560 [ACK] Seq=A Ack= B Win=C Len=0

Figure 2: Screenshot from Wireshark.

- (a) Identify all TCP buffers involved in data exchange and, identify which buffer(s) the TCP Window mechanism relates to. (5 marks)
- (b) Explain the operation of the TCP Window mechanism during data exchange. In your answer, explain how the value in the Win field fluctuates during data exchange and, the consequence for both the sender and receiver TCP entities should this field contain a zero value. (8 marks)
- (c) With reference to lines 4 in figure 2 above, the **total** length of the associated segment is 477 and the segment contains 20 bytes of **header** data. Identify the missing values A, B & C in the three fields of the ACK segments returned on line 5. For each field, justify your answer with reference to values contained in previous segment transmissions. (12 marks)